

Лабораторная работа №7

Управление журналами событий в системе

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Цель работы

Получить навыки работы с журналами мониторинга различных событий в системе Linux.

Ход выполнения работы

```
root@rishardkogengar:/home/rkogengar# tail -f /var/log/messages
Sep 29 13:11:59 rishardkogengar kernel: traps: VBoxClient[3683] trap int3 ip:41dd1b sp:7fed3fc35cd
0 error:0 in VBoxClient[1dd1b,400000+bb000]
Sep 29 13:11:59 rishardkogengar systemd-coredump[3684]: Process 3680 (VBoxClient) of user 1000 ter
minated abnormally with signal 5/TRAP, processing...
Sep 29 13:11:59 rishardkogengar systemd[1]: Started systemd-coredump@35-3684-0.service - Process C
ore Dump (PID 3684/UID 0).
Sep 29 13:11:59 rishardkogengar systemd-coredump[3685]: Process 3680 (VBoxClient) of user 1000 dum
ped core.#012#012Module libXau.so.6 from rpm libXau-1.0.11-8.el10.x86_64#012Module libxcb.so.1 fro
m rpm libxcb-1.17.0-3.el10.x86_64#012Module libX11.so.6 from rpm libX11-1.8.10-1.el10.x86_64#012Mo
dule libffi.so.8 from rpm libffi-3.4.4-9.el10.x86_64#012Module libwayland-client.so.0 from rpm way
land-1.23.0-2.el10.x86_64#012Stack trace of thread 3683:#012#0 0x000000000041dd1b n/a (n/a + 0x0)
#012#1 0x000000000041dc94 n/a (n/a + 0x0)#012#2 0x000000000045041c n/a (n/a + 0x0)#012#3 0x0000
0000004355d0 n/a (n/a + 0x0)#012#4 0x00007fed4e32b11a start_thread (libc.so.6 + 0x9511a)#012#5 0
x00007fed4e39bc3c __clone3 (libc.so.6 + 0x105c3c)#012#012Stack trace of thread 3680:#012#0 0x0000
7fed4e399a3d syscall (libc.so.6 + 0x103a3d)#012#1 0x00000000004344e2 n/a (n/a + 0x0)#012#2 0x000
0000000450066 n/a (n/a + 0x0)#012#3 0x0000000000405123 n/a (n/a + 0x0)#012#4 0x00007fed4e2c030e
__libc_start_call_main (libc.so.6 + 0x2a30e)#012#5 0x00007fed4e2c03c9 __libc_start_main@@GLIBC_2.
34 (libc.so.6 + 0x2a3c9)#012#6 0x00000000004044aa n/a (n/a + 0x0)#012ELF object binary architectu
re: AMD x86-64
Sep 29 13:11:59 rishardkogengar systemd[1]: systemd-coredump@35-3684-0.service: Deactivated succes
```

Рис. 1: Мониторинг системных событий

```
34 (libc.so.0 + 0x2a3c9)#012#0 0x0000000000004044aa n/a (n/a + 0x0)#012ELF object binary architecture: AMD x86-64
Sep 29 13:12:25 rishardkogengar systemd[1]: systemd-coredump@40-3748-0.service: Deactivated successfully.
Sep 29 13:12:25 rishardkogengar systemd[1]: Starting fprintd.service - Fingerprint Authentication Daemon...
Sep 29 13:12:25 rishardkogengar systemd[1]: Started fprintd.service - Fingerprint Authentication Daemon.
Sep 29 13:12:29 rishardkogengar su[3754]: FAILED SU (to root) rkogengar on pts/2
Sep 29 13:12:30 rishardkogengar kernel: traps: VBoxClient[3767] trap int3 ip:41dd1b sp:7fed3fc35cd0 error:0 in VBoxClient[1dd1b,400000+bb000]
Sep 29 13:12:30 rishardkogengar systemd-coredump[3768]: Process 3764 (VBoxClient) of user 1000 terminated abnormally with signal 5/TRAP, processing...
Sep 29 13:12:30 rishardkogengar systemd[1]: Started systemd-coredump@41-3768-0.service - Process C
```

Рис. 2: Ошибка авторизации su

```
34 (libc.so.6 + 0x2a3c9)#012#6 0x00000000004044aa n/a (n/a + 0x0)#012ELF object binary architecture: AMD x86-64
Sep 29 13:12:50 rishardkogengar systemd[1]: systemd-coredump@45-3810-0.service: Deactivated successfully.
Sep 29 13:12:52 rishardkogengar rkogengar[3816]: hello
Sep 29 13:12:55 rishardkogengar kernel: traps: VBoxClient[3824] trap int3 ip:41dd1b sp:7fed3fc35cd 0 error:0 in VBoxClient[1dd1b,400000+bb000]
Sep 29 13:12:55 rishardkogengar systemd-coredump[3825]: Process 3821 (VBoxClient) of user 1000 terminated abnormally with signal 5/TRAP, processing...
Sep 29 13:12:55 rishardkogengar systemd[1]: Started systemd-coredump@46-3825-0.service - Process Core Dump (PID 3825/UID 0).
```

Рис. 3: Сообщение от logger

```
root@rishardkogengar:/home/rkogengar# tail -n 20 /var/log/secure
Sep 29 13:08:12 rishardkogengar sshd[1191]: Server listening on :: port 22.
Sep 29 13:08:12 rishardkogengar (systemd)[1259]: pam_unix(systemd-user:session): session opened for user gdm(uid=42) by gdm(uid=0)
Sep 29 13:08:12 rishardkogengar gdm-launch-environment[1235]: pam_unix(gdm-launch-environment:session): session opened for user gdm(uid=42) by (uid=0)
Sep 29 13:08:44 rishardkogengar unix_chkpwd[1991]: password check failed for user (rkogengar)
Sep 29 13:08:44 rishardkogengar gdm-password[1983]: pam_unix(gdm-password:auth): authentication failure; logname= uid=0 euid=0 tty=/dev/tty1 ruser= rhost= user=rkogengar
Sep 29 13:08:44 rishardkogengar gdm-password[1983]: gkr-pam: unable to locate daemon control file
Sep 29 13:08:44 rishardkogengar gdm-password[1983]: gkr-pam: stashed password to try later in open session
Sep 29 13:08:49 rishardkogengar gdm-password[1994]: gkr-pam: unable to locate daemon control file
Sep 29 13:08:49 rishardkogengar gdm-password[1994]: gkr-pam: stashed password to try later in open session
Sep 29 13:08:50 rishardkogengar (systemd)[2004]: pam_unix(systemd-user:session): session opened for user rkogengar(uid=1000) by rkogengar(uid=0)
Sep 29 13:08:50 rishardkogengar gdm-password[1994]: pam_unix(gdm-password:session): session opened for user rkogengar(uid=1000) by rkogengar(uid=0)
Sep 29 13:08:50 rishardkogengar gdm-password[1994]: gkr-pam: gnome-keyring-daemon started properly and unlocked keyring
Sep 29 13:09:00 rishardkogengar gdm-launch-environment[1235]: pam_unix(gdm-launch-environment:session): session closed for user gdm
Sep 29 13:11:41 rishardkogengar (systemd)[3497]: pam_unix(systemd-user:session): session opened for user root(uid=0) by root(uid=0)
Sep 29 13:11:42 rishardkogengar su[3483]: pam_unix(su:session): session opened for user root(uid=0) by rkogengar(uid=1000)
Sep 29 13:11:49 rishardkogengar su[3577]: pam_unix(su:session): session opened for user root(uid=0)
```

Рис. 4: Просмотр последних строк secure

Изменение правил rsyslog.conf

```
Installed:
  apr-1.7.5-2.el10.x86_64
  apr-util-lmdb-1.6.3-21.el10.x86_64
  httpd-2.4.63-1.el10_0.2.x86_64
  httpd-filesystem-2.4.63-1.el10_0.2.noarch
  mod_http2-2.0.29-2.el10_0.1.x86_64
  rocky-logos-httpd-100.4-7.el10.noarch
  apr-util-1.6.3-21.el10.x86_64
  apr-util-openssl-1.6.3-21.el10.x86_64
  httpd-core-2.4.63-1.el10_0.2.x86_64
  httpd-tools-2.4.63-1.el10_0.2.x86_64
  mod_lua-2.4.63-1.el10_0.2.x86_64

Complete!
root@rishardkogengar:/home/rkogengar# systemctl start httpd
root@rishardkogengar:/home/rkogengar# systemctl enable httpd
Created symlink '/etc/systemd/system/multi-user.target.wants/httpd.service' → '/usr/lib/systemd/system/httpd.service'.
root@rishardkogengar:/home/rkogengar#
```

Рис. 5: Установка и запуск httpd

```
root@rishardkogengar:/home/rkogengar#
root@rishardkogengar:/home/rkogengar#
root@rishardkogengar:/home/rkogengar# tail -f /var/log/httpd/error_log
[Mon Sep 29 13:15:47.008133 2025] [suexec:notice] [pid 4444:tid 4444] AH01232: suEXEC mechanism enabled (wrapper: /usr/sbin/suexec)
[Mon Sep 29 13:15:47.043951 2025] [lbmethod_heartbeat:notice] [pid 4444:tid 4444] AH02282: No slot mem from mod_heartbeat
[Mon Sep 29 13:15:47.044511 2025] [systemd:notice] [pid 4444:tid 4444] SELinux policy enabled; httpd running as context system_u:system_r:httpd_t:s0
[Mon Sep 29 13:15:47.045975 2025] [mpm_event:notice] [pid 4444:tid 4444] AH00489: Apache/2.4.63 (Rocky Linux) configured -- resuming normal operations
[Mon Sep 29 13:15:47.046012 2025] [core:notice] [pid 4444:tid 4444] AH00094: Command line: '/usr/sbin/httpd -D FOREGROUND'
```

Использование journalctl

```
root@rishardkogengar: /home/rkogengar# journalctl
root@rishardkogengar: /home/rkogengar# journalctl
Sep 29 13:08:06 rishardkogengar.localdomain kernel: Linux version 6.12.0-55.12.1.el10_0.x86_64 (m
Sep 29 13:08:06 rishardkogengar.localdomain kernel: Command line: BOOT_IMAGE=(hd0,gpt2)/vmlinuz-6
Sep 29 13:08:06 rishardkogengar.localdomain kernel: BIOS-provided physical RAM map:
Sep 29 13:08:06 rishardkogengar.localdomain kernel: BIOS-e820: [mem 0x0000000000000000-0x00000000
Sep 29 13:08:06 rishardkogengar.localdomain kernel: BIOS-e820: [mem 0x0000000000009fc00-0x00000000
Sep 29 13:08:06 rishardkogengar.localdomain kernel: BIOS-e820: [mem 0x000000000000f0000-0x00000000
Sep 29 13:08:06 rishardkogengar.localdomain kernel: BIOS-e820: [mem 0x00000000000100000-0x00000000
Sep 29 13:08:06 rishardkogengar.localdomain kernel: BIOS-e820: [mem 0x0000000000dffff0000-0x00000000
Sep 29 13:08:06 rishardkogengar.localdomain kernel: BIOS-e820: [mem 0x000000000fec00000-0x00000000
Sep 29 13:08:06 rishardkogengar.localdomain kernel: BIOS-e820: [mem 0x00000000fee00000-0x00000000
Sep 29 13:08:06 rishardkogengar.localdomain kernel: BIOS-e820: [mem 0x00000000fffc0000-0x00000000
Sep 29 13:08:06 rishardkogengar.localdomain kernel: BIOS-e820: [mem 0x00000000100000000-0x00000001
Sep 29 13:08:06 rishardkogengar.localdomain kernel: NX (Execute Disable) protection: active
Sep 29 13:08:06 rishardkogengar.localdomain kernel: APIC: Static calls initialized
Sep 29 13:08:06 rishardkogengar.localdomain kernel: SMBIOS 2.5 present.
Sep 29 13:08:06 rishardkogengar.localdomain kernel: DMI: innotek GmbH VirtualBox/VirtualBox, BIOS
Sep 29 13:08:06 rishardkogengar.localdomain kernel: DMI: Memory slots populated: 0/0
Sep 29 13:08:06 rishardkogengar.localdomain kernel: Hypervisor detected: KVM
Sep 29 13:08:06 rishardkogengar.localdomain kernel: kvm-clock: Using msrs 4b564d01 and 4b564d00
Sep 29 13:08:06 rishardkogengar.localdomain kernel: kvm-clock: using sched offset of 4056055671 c
Sep 29 13:08:06 rishardkogengar.localdomain kernel: clocksource: kvm-clock: mask: 0xffffffffffff
Sep 29 13:08:06 rishardkogengar.localdomain kernel: tsc: Detected 3187.202 MHz processor
Sep 29 13:08:06 rishardkogengar.localdomain kernel: e820: update [mem 0x00000000-0x00000fff] usab
Sep 29 13:08:06 rishardkogengar.localdomain kernel: e820: remove [mem 0x000a0000-0x000fffff] usab
Sep 29 13:08:06 rishardkogengar.localdomain kernel: last_pfn = 0x120000 max_arch_pfn = 0x40000000
Sep 29 13:08:06 rishardkogengar.localdomain kernel: total RAM covered: 4096M
```

Рис. 11: Просмотр системного журнала

Использование journalctl

```
Sep 29 13:25:33 rishardkogengar.localdomain systemd[1]: Started systemd-coredump@194-6429-0.servic
e - Process Core Dump (PID 6429/UID 0).
Sep 29 13:25:33 rishardkogengar.localdomain systemd-coredump[6430]: [?] Process 6425 (VBoxClient)
of user 1000 dumped core.

                                Module libXau.so.6 from rpm li
bXau-1.0.11-8.el10.x86_64
                                Module libxcb.so.1 from rpm li
bxcb-1.17.0-3.el10.x86_64
                                Module libX11.so.6 from rpm li
bX11-1.8.10-1.el10.x86_64
                                Module libffi.so.8 from rpm li
bffi-3.4.4-9.el10.x86_64
                                Module libwayland-client.so.0
from rpm wayland-1.23.0-2.el10.x86_64

                                Stack trace of thread 6428:
#0  0x00000000041dd1b n/a (n/
#1  0x00000000041dc94 n/a (n/
#2  0x00000000045041c n/a (n/
#3  0x0000000004355d0 n/a (n/
#4  0x00007fed4e32b11a start_t
#5  0x00007fed4e39bc3c __clone

                                Stack trace of thread 6427:
```

Рис. 12: journalctl без пейджера

Использование journalctl

```
from rpm wayland-1.23.0-2.el10.x86_64

a + 0x0)

a + 0x0)

a + 0x0)

a + 0x0)

hread (libc.so.6 + 0x9511a)

3 (libc.so.6 + 0x105c3c)

(libc.so.6 + 0x103a3d)

a + 0x0)

a + 0x0)

a + 0x0)

start_call_main (libc.so.6 + 0x2a30e)

Module libwayland-client.so.0

Stack trace of thread 6527:
#0  0x000000000041dd1b n/a (n/
#1  0x000000000041dc94 n/a (n/
#2  0x000000000045041c n/a (n/
#3  0x00000000004355d0 n/a (n/
#4  0x00007fed4e32b11a start_t
#5  0x00007fed4e39bc3c __clone

Stack trace of thread 6524:
#0  0x00007fed4e399a3d syscall
#1  0x00000000004344e2 n/a (n/
#2  0x0000000000450066 n/a (n/
#3  0x0000000000405123 n/a (n/
#4  0x00007fed4e2c030e __libc_
#5  0x00007fed4e2c03c9 __libc_
```

Рис. 13: journalctl в реальном времени

Использование journalctl

```
root@rishardkogengar: /home/rkogengar# journalctl
Display all 128 possibilities? (y or n)
_AUDIT_LOGINUID=                                JOB_TYPE=
_AUDIT_SESSION=                                JOURNAL_NAME=
AVAILABLE=                                       JOURNAL_PATH=
AVAILABLE_PRETTY=                              _KERNEL_DEVICE=
_BOOT_ID=                                       _KERNEL_SUBSYSTEM=
_CAP_EFFECTIVE=                                KERNEL_USEC=
_CMDLINE=                                       LEADER=
CODE_FILE=                                     LIMIT=
CODE_FUNC=                                     LIMIT_PRETTY=
CODE_LINE=                                     _LINE_BREAK=
_COMM=                                         _MACHINE_ID=
CONFIG_FILE=                                   MAX_USE=
CONFIG_LINE=                                   MAX_USE_PRETTY=
COREDUMP_CGROUP=                               MEMORY_PEAK=
COREDUMP_CMDLINE=                             MEMORY_SWAP_PEAK=
COREDUMP_COMM=                                 MESSAGE=
COREDUMP_CWD=                                 MESSAGE_ID=
COREDUMP_ENVIRON=                             NM_DEVICE=
COREDUMP_EXE=                                 NM_LOG_DOMAINS=
COREDUMP_FILENAME=                           NM_LOG_LEVEL=
COREDUMP_GID=                                 _PID=
COREDUMP_HOSTNAME=                           PODMAN_EVENT=
COREDUMP_OPEN_FDS=                           PODMAN_TIME=
COREDUMP_OWNER_UID=                           PODMAN_TYPE=
COREDUMP_PACKAGE_JSON=                       PRIORITY=
COREDUMP_PID=                                 REALMD_OPERATION=
COREDUMP_PROC_AUXV=                           _RUNTIME_SCOPE=
COREDUMP_PROC_CGROUP=                         SEAT_ID=
COREDUMP_PROC_LIMITS=                         _SELINUX_CONTEXT=
COREDUMP_PROC_MAPS=                           SESSION_ID=
COREDUMP_PROC_MOUNTINFO=                     _SOURCE_BOOTTIME_TIMESTAMP=
```

Использование journalctl

```
root@rishardkogengar:/home/rkogengar#  
root@rishardkogengar:/home/rkogengar# journalctl _UID=0  
Sep 29 13:08:06 rishardkogengar.localdomain systemd-journald[284]: Collecting audit messages is disabled  
Sep 29 13:08:06 rishardkogengar.localdomain systemd-journald[284]: Journal started  
Sep 29 13:08:06 rishardkogengar.localdomain systemd-journald[284]: Runtime Journal (/run/log/journal/rishardkogengar.localdomain) is empty  
Sep 29 13:08:06 rishardkogengar.localdomain systemd-modules-load[285]: Module 'msr' is built in  
Sep 29 13:08:06 rishardkogengar.localdomain systemd-modules-load[285]: Inserted module 'fuse'  
Sep 29 13:08:06 rishardkogengar.localdomain systemd-modules-load[285]: Module 'scsi_dh_alua' is built in  
Sep 29 13:08:06 rishardkogengar.localdomain systemd-modules-load[285]: Module 'scsi_dh_emc' is built in  
Sep 29 13:08:06 rishardkogengar.localdomain systemd-modules-load[285]: Module 'scsi_dh_rdac' is built in  
Sep 29 13:08:06 rishardkogengar.localdomain systemd-sysusers[295]: Creating group 'nobody' with GID 4294967295  
Sep 29 13:08:06 rishardkogengar.localdomain systemd[1]: Finished systemd-sysctl.service - Apply kernel parameters  
Sep 29 13:08:06 rishardkogengar.localdomain systemd-sysusers[295]: Creating group 'users' with GID 1000  
Sep 29 13:08:06 rishardkogengar.localdomain systemd-sysusers[295]: Creating group 'systemd-journal' with GID 1000  
Sep 29 13:08:06 rishardkogengar.localdomain systemd[1]: Finished systemd-sysusers.service - Create system users and groups  
Sep 29 13:08:06 rishardkogengar.localdomain systemd[1]: Starting systemd-tmpfiles-setup-dev.service - Create and update temporary files and directories  
Sep 29 13:08:07 rishardkogengar.localdomain systemd[1]: Finished systemd-vconsole-setup.service - Set up vconsole  
Sep 29 13:08:07 rishardkogengar.localdomain systemd[1]: dracut-cmdline-ask.service - dracut ask for kernel command line parameters  
Sep 29 13:08:07 rishardkogengar.localdomain systemd[1]: Starting dracut-cmdline.service - dracut cmdline generator  
Sep 29 13:08:07 rishardkogengar.localdomain dracut-cmdline[311]: dracut-105-4.el10_0  
Sep 29 13:08:07 rishardkogengar.localdomain dracut-cmdline[311]: Using kernel command line parameters  
Sep 29 13:08:07 rishardkogengar.localdomain systemd[1]: Finished systemd-tmpfiles-setup-dev.service - Create and update temporary files and directories  
Sep 29 13:08:07 rishardkogengar.localdomain systemd[1]: Finished dracut-cmdline.service - dracut cmdline generator  
Sep 29 13:08:07 rishardkogengar.localdomain systemd[1]: Starting dracut-pre-udev.service - dracut pre-udev generator  
Sep 29 13:08:07 rishardkogengar.localdomain systemd[1]: Finished dracut-pre-udev.service - dracut pre-udev generator  
Sep 29 13:08:07 rishardkogengar.localdomain systemd[1]: Starting systemd-udevd.service - Rule-based device manager  
Sep 29 13:08:07 rishardkogengar.localdomain systemd-udevd[409]: Using default interface naming scheme 'vusb' for new devices  
Sep 29 13:08:07 rishardkogengar.localdomain systemd[1]: Started systemd-udevd.service - Rule-based device manager
```

Рис. 15: События UID=0

Использование journalctl

```
root@rishardkogengar:/home/ikogengar# journalctl -n 20
Sep 29 13:27:15 rishardkogengar.localdomain systemd[1]: Started systemd-coredump@214-6728-0.servi>
Sep 29 13:27:15 rishardkogengar.localdomain systemd-coredump[6729]: [...] Process 6724 (VBoxClient)>
Module libXau.so.6 from rpm l>
Module libxcb.so.1 from rpm l>
Module libX11.so.6 from rpm l>
Module libffi.so.8 from rpm l>
Module libwayland-client.so.0>
Stack trace of thread 6727:
#0 0x00000000041dd1b n/a (n>
#1 0x00000000041dc94 n/a (n>
#2 0x00000000045041c n/a (n>
#3 0x0000000004355d0 n/a (n>
#4 0x00007fed4e32b11a start_>
#5 0x00007fed4e39bc3c __clon>
Stack trace of thread 6725:
#0 0x00007fed4e399a3d syscal>
#1 0x000000000434c30 n/a (n>
#2 0x000000000450bfb n/a (n>
#3 0x00000000043566a n/a (n>
#4 0x00000000045041c n/a (n>
```

Рис. 16: Последние 20 строк журнала

Использование journalctl

```
root@rishardkogengar: /home/rkogengar#  
root@rishardkogengar: /home/rkogengar# journalctl -p err  
Sep 29 13:08:07 rishardkogengar.localdomain kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx see  
Sep 29 13:08:07 rishardkogengar.localdomain kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This confi  
Sep 29 13:08:07 rishardkogengar.localdomain kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please swi  
Sep 29 13:08:10 rishardkogengar.localdomain kernel: Warning: Unmaintained driver is detected: e10  
Sep 29 13:08:11 rishardkogengar.localdomain alsactl[920]: alsa-lib main.c:1554:(snd_use_case_mgr_>  
Sep 29 13:08:12 rishardkogengar.localdomain kernel: Warning: Unmaintained driver is detected: ip_>  
Sep 29 13:08:44 rishardkogengar.localdomain gdm-password[1983]: gkr-pam: unable to locate daemon>  
Sep 29 13:08:49 rishardkogengar.localdomain gdm-password[1994]: gkr-pam: unable to locate daemon>  
Sep 29 13:08:59 rishardkogengar.localdomain systemd[2004]: Failed to start app-gnome-gnome\x2dkey>  
Sep 29 13:09:01 rishardkogengar.localdomain systemd-coredump[2813]: [..] Process 2796 (VBoxClient)>
```

```
Module libXau.so.6 from rpm l  
Module libxcb.so.1 from rpm l  
Module libX11.so.6 from rpm l  
Module libffi.so.8 from rpm l  
Module libwayland-client.so.0>
```

Stack trace of thread 2803:

```
#0 0x000000000041dd1b n/a (n  
#1 0x000000000041dc94 n/a (n  
#2 0x000000000045041c n/a (n  
#3 0x00000000004355d0 n/a (n  
#4 0x00007fed4e32b11a start_>  
#5 0x00007fed4e39bc3c __clon>
```

Stack trace of thread 2800:

```
#0 0x00007fed4e399a3d syscal>  
#1 0x00000000004344e2 n/a (n  
#2 0x0000000000450066 n/a (n  
#3 0x000000000041c550 n/a (n
```

Рис. 17: Сообщения об ошибках

Использование journalctl

```
root@rishardkogengar:/home/rkogengar# journalctl --since yesterday
Sep 29 13:08:06 rishardkogengar.localdomain kernel: Linux version 6.12.0-55.12.1.el10_0.x86_64 (m
Sep 29 13:08:06 rishardkogengar.localdomain kernel: Command line: BOOT_IMAGE=(hd0,gpt2)/vmlinuz-6
Sep 29 13:08:06 rishardkogengar.localdomain kernel: BIOS-provided physical RAM map:
Sep 29 13:08:06 rishardkogengar.localdomain kernel: BIOS-e820: [mem 0x0000000000000000-0x00000000
Sep 29 13:08:06 rishardkogengar.localdomain kernel: BIOS-e820: [mem 0x00000000000009fc00-0x00000000
Sep 29 13:08:06 rishardkogengar.localdomain kernel: BIOS-e820: [mem 0x000000000000f0000-0x00000000
Sep 29 13:08:06 rishardkogengar.localdomain kernel: BIOS-e820: [mem 0x0000000000100000-0x00000000
Sep 29 13:08:06 rishardkogengar.localdomain kernel: BIOS-e820: [mem 0x000000000dffff0000-0x00000000
Sep 29 13:08:06 rishardkogengar.localdomain kernel: BIOS-e820: [mem 0x00000000fec00000-0x00000000
Sep 29 13:08:06 rishardkogengar.localdomain kernel: BIOS-e820: [mem 0x00000000fee00000-0x00000000
Sep 29 13:08:06 rishardkogengar.localdomain kernel: BIOS-e820: [mem 0x00000000fffc0000-0x00000000
Sep 29 13:08:06 rishardkogengar.localdomain kernel: BIOS-e820: [mem 0x00000000100000000-0x00000001
Sep 29 13:08:06 rishardkogengar.localdomain kernel: NX (Execute Disable) protection: active
Sep 29 13:08:06 rishardkogengar.localdomain kernel: APIC: Static calls initialized
Sep 29 13:08:06 rishardkogengar.localdomain kernel: SMBIOS 2.5 present.
Sep 29 13:08:06 rishardkogengar.localdomain kernel: DMI: innotek GmbH VirtualBox/VirtualBox, BIOS
Sep 29 13:08:06 rishardkogengar.localdomain kernel: DMI: Memory slots populated: 0/0
Sep 29 13:08:06 rishardkogengar.localdomain kernel: Hypervisor detected: KVM
Sep 29 13:08:06 rishardkogengar.localdomain kernel: kvm-clock: Using msrs 4b564d01 and 4b564d00
Sep 29 13:08:06 rishardkogengar.localdomain kernel: kvm-clock: using sched offset of 4056055671 c
Sep 29 13:08:06 rishardkogengar.localdomain kernel: clocksource: kvm-clock: mask: 0xffffffffffff
Sep 29 13:08:06 rishardkogengar.localdomain kernel: tsc: Detected 3187.202 MHz processor
Sep 29 13:08:06 rishardkogengar.localdomain kernel: e820: update [mem 0x00000000-0x00000fff] usab
Sep 29 13:08:06 rishardkogengar.localdomain kernel: e820: remove [mem 0x000a0000-0x000fffff] usab
Sep 29 13:08:06 rishardkogengar.localdomain kernel: last_pfn = 0x120000 max_arch_pfn = 0x400000000
Sep 29 13:08:06 rishardkogengar.localdomain kernel: total RAM covered: 4096M
Sep 29 13:08:06 rishardkogengar.localdomain kernel: Found optimal setting for mtrr clean up
```

Рис. 18: Журнал с определённой даты

Использование journalctl

```
root@rishardkogengar:/home/rkogengar#
root@rishardkogengar:/home/rkogengar# journalctl --since yesterday -p err
Sep 29 13:08:07 rishardkogengar.localdomain kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx see>
Sep 29 13:08:07 rishardkogengar.localdomain kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This confi>
Sep 29 13:08:07 rishardkogengar.localdomain kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please swi>
Sep 29 13:08:10 rishardkogengar.localdomain kernel: Warning: Unmaintained driver is detected: e10>
Sep 29 13:08:11 rishardkogengar.localdomain alsactl[920]: alsa-lib main.c:1554:(snd_use_case_mgr>
Sep 29 13:08:12 rishardkogengar.localdomain kernel: Warning: Unmaintained driver is detected: ip>
Sep 29 13:08:44 rishardkogengar.localdomain gdm-password[1983]: gkr-pam: unable to locate daemon>
Sep 29 13:08:49 rishardkogengar.localdomain gdm-password[1994]: gkr-pam: unable to locate daemon>
Sep 29 13:08:59 rishardkogengar.localdomain systemd[2004]: Failed to start app-gnome-gnome\x2dkey>
Sep 29 13:09:01 rishardkogengar.localdomain systemd-coredump[2813]: [?] Process 2796 (VBoxClient)>

Module libXau.so.6 from rpm l>
Module libxcb.so.1 from rpm l>
Module libX11.so.6 from rpm l>
Module libffi.so.8 from rpm l>
Module libwayland-client.so.0>
Stack trace of thread 2803:
#0 0x000000000041dd1b n/a (n>
#1 0x000000000041dc94 n/a (n>
#2 0x000000000045041c n/a (n>
#3 0x00000000004355d0 n/a (n>
#4 0x00007fed4e32b11a start>
#5 0x00007fed4e39bc3c __clon>

Stack trace of thread 2800:
#0 0x00007fed4e399a3d syscal>
#1 0x00000000004344e2 n/a (n>
#2 0x0000000000450066 n/a (n>
```

Рис. 19: Ошибки со вчерашнего дня

Использование journalctl

```
-----  
Mon 2025-09-29 13:08:06.926590 MSK [s=5f741529396a49ccbdded82bc18133f9b;i=2;b=e412ee26822c4f7cba16>  
_SOURCE_BOOTTIME_TIMESTAMP=0  
_SOURCE_MONOTONIC_TIMESTAMP=0  
_TRANSPORT=kernel  
SYSLOG_FACILITY=0  
SYSLOG_IDENTIFIER=kernel  
_BOOT_ID=e412ee26822c4f7cba161f7de6ecee11  
_MACHINE_ID=9b971042516b4d0cb2d9a4048b7180f9  
_HOSTNAME=rishardkogengar.localdomain  
_RUNTIME_SCOPE=initrd  
PRIORITY=6  
MESSAGE=Command line: BOOT_IMAGE=(hd0,gpt2)/vmlinuz-6.12.0-55.12.1.el10_0.x86_64 root=/dev/ma>  
Mon 2025-09-29 13:08:06.926595 MSK [s=5f741529396a49ccbdded82bc18133f9b;i=3;b=e412ee26822c4f7cba16>  
_SOURCE_BOOTTIME_TIMESTAMP=0  
_SOURCE_MONOTONIC_TIMESTAMP=0  
_TRANSPORT=kernel  
SYSLOG_FACILITY=0  
SYSLOG_IDENTIFIER=kernel  
_BOOT_ID=e412ee26822c4f7cba161f7de6ecee11  
_MACHINE_ID=9b971042516b4d0cb2d9a4048b7180f9  
_HOSTNAME=rishardkogengar.localdomain  
_RUNTIME_SCOPE=initrd  
PRIORITY=6  
root@rishardkogengar:/home/rkogengar# journalctl _SYSTEMD_UNIT=sshd.service  
Sep 29 13:08:12 rishardkogengar.localdomain (sshd)[1191]: sshd.service: Referenced but unset envt>  
Sep 29 13:08:12 rishardkogengar.localdomain sshd[1191]: Server listening on 0.0.0.0 port 22.  
Sep 29 13:08:12 rishardkogengar.localdomain sshd[1191]: Server listening on :: port 22.  
lines 1-3/3 (END)
```

Рис. 20: Детализированный вывод verbose

Использование journalctl

```
root@rishardkogengar:/home/rkogengar#
root@rishardkogengar:/home/rkogengar# mkdir -p /var/log/journal
root@rishardkogengar:/home/rkogengar# chown root:systemd-journal
chown: missing operand after 'root:systemd-journal'
Try 'chown --help' for more information.
root@rishardkogengar:/home/rkogengar# chown root:systemd-journal /var/log/journal/
root@rishardkogengar:/home/rkogengar# chmod 2755 /var/log/journal/
root@rishardkogengar:/home/rkogengar# killall -USR1 systemd-journald
root@rishardkogengar:/home/rkogengar# journalctl -b
Sep 29 13:08:06 rishardkogengar.localdomain kernel: Linux version 6.12.0-55.12.1.el10_0.x86_64 (m
Sep 29 13:08:06 rishardkogengar.localdomain kernel: Command line: BOOT_IMAGE=(hd0,gpt2)/vmlinuz-6>
Sep 29 13:08:06 rishardkogengar.localdomain kernel: BIOS-provided physical RAM map:
Sep 29 13:08:06 rishardkogengar.localdomain kernel: BIOS-e820: [mem 0x0000000000000000-0x00000000>
Sep 29 13:08:06 rishardkogengar.localdomain kernel: BIOS-e820: [mem 0x0000000000009fc00-0x00000000>
Sep 29 13:08:06 rishardkogengar.localdomain kernel: BIOS-e820: [mem 0x000000000000f0000-0x00000000>
Sep 29 13:08:06 rishardkogengar.localdomain kernel: BIOS-e820: [mem 0x00000000000100000-0x00000000>
Sep 29 13:08:06 rishardkogengar.localdomain kernel: BIOS-e820: [mem 0x0000000000dffff0000-0x00000000>
Sep 29 13:08:06 rishardkogengar.localdomain kernel: BIOS-e820: [mem 0x0000000000fec000000-0x00000000>
Sep 29 13:08:06 rishardkogengar.localdomain kernel: BIOS-e820: [mem 0x0000000000fee000000-0x00000000>
Sep 29 13:08:06 rishardkogengar.localdomain kernel: BIOS-e820: [mem 0x0000000000fffc00000-0x00000000>
Sep 29 13:08:06 rishardkogengar.localdomain kernel: BIOS-e820: [mem 0x0000000001000000000-0x000000001>
Sep 29 13:08:06 rishardkogengar.localdomain kernel: NX (Execute Disable) protection: active
Sep 29 13:08:06 rishardkogengar.localdomain kernel: APIC: Static calls initialized
Sep 29 13:08:06 rishardkogengar.localdomain kernel: SMBIOS 2.5 present.
Sep 29 13:08:06 rishardkogengar.localdomain kernel: DMI: innotek GmbH VirtualBox/VirtualBox, BIOS>
Sep 29 13:08:06 rishardkogengar.localdomain kernel: DMI: Memory slots populated: 0/0
Sep 29 13:08:06 rishardkogengar.localdomain kernel: Hypervisor detected: KVM
Sep 29 13:08:06 rishardkogengar.localdomain kernel: kvm-clock: Using msrs 4b564d01 and 4b564d00
Sep 29 13:08:06 rishardkogengar.localdomain kernel: kvm-clock: using sched offset of 4056055671 c>
Sep 29 13:08:06 rishardkogengar.localdomain kernel: clocksource: kvm-clock: mask: 0xffffffffffff>
Sep 29 13:08:06 rishardkogengar.localdomain kernel: tsc: Detected 3187.202 MHz processor
Sep 29 13:08:06 rishardkogengar.localdomain kernel: e820: update [mem 0x000000000-0x000000fff] usab>
```

Постоянный журнал journald

```
root@rishardkogengar:/home/rkogengar#
root@rishardkogengar:/home/rkogengar# mkdir -p /var/log/journal
root@rishardkogengar:/home/rkogengar# chown root:systemd-journal
chown: missing operand after 'root:systemd-journal'
Try 'chown --help' for more information.
root@rishardkogengar:/home/rkogengar# chown root:systemd-journal /var/log/journal/
root@rishardkogengar:/home/rkogengar# chmod 2755 /var/log/journal/
root@rishardkogengar:/home/rkogengar# killall -USR1 systemd-journald
root@rishardkogengar:/home/rkogengar# journalctl -b
Sep 29 13:08:06 rishardkogengar.localdomain kernel: Linux version 6.12.0-55.12.1.el10_0.x86_64 (m
Sep 29 13:08:06 rishardkogengar.localdomain kernel: Command line: BOOT_IMAGE=(hd0,gpt2)/vmlinuz-6>
Sep 29 13:08:06 rishardkogengar.localdomain kernel: BIOS-provided physical RAM map:
Sep 29 13:08:06 rishardkogengar.localdomain kernel: BIOS-e820: [mem 0x0000000000000000-0x00000000>
Sep 29 13:08:06 rishardkogengar.localdomain kernel: BIOS-e820: [mem 0x0000000000009fc00-0x00000000>
Sep 29 13:08:06 rishardkogengar.localdomain kernel: BIOS-e820: [mem 0x000000000000f0000-0x00000000>
Sep 29 13:08:06 rishardkogengar.localdomain kernel: BIOS-e820: [mem 0x00000000000100000-0x00000000>
Sep 29 13:08:06 rishardkogengar.localdomain kernel: BIOS-e820: [mem 0x0000000000dffff0000-0x00000000>
Sep 29 13:08:06 rishardkogengar.localdomain kernel: BIOS-e820: [mem 0x000000000fec00000-0x00000000>
Sep 29 13:08:06 rishardkogengar.localdomain kernel: BIOS-e820: [mem 0x000000000fee00000-0x00000000>
Sep 29 13:08:06 rishardkogengar.localdomain kernel: BIOS-e820: [mem 0x000000000fffc0000-0x00000000>
Sep 29 13:08:06 rishardkogengar.localdomain kernel: BIOS-e820: [mem 0x00000000100000000-0x000000001>
Sep 29 13:08:06 rishardkogengar.localdomain kernel: NX (Execute Disable) protection: active
Sep 29 13:08:06 rishardkogengar.localdomain kernel: APIC: Static calls initialized
Sep 29 13:08:06 rishardkogengar.localdomain kernel: SMBIOS 2.5 present.
Sep 29 13:08:06 rishardkogengar.localdomain kernel: DMI: innotek GmbH VirtualBox/VirtualBox, BIOS>
Sep 29 13:08:06 rishardkogengar.localdomain kernel: DMI: Memory slots populated: 0/0
Sep 29 13:08:06 rishardkogengar.localdomain kernel: Hypervisor detected: KVM
Sep 29 13:08:06 rishardkogengar.localdomain kernel: kvm-clock: Using msrs 4b564d01 and 4b564d00
Sep 29 13:08:06 rishardkogengar.localdomain kernel: kvm-clock: using sched offset of 4056055671 c>
Sep 29 13:08:06 rishardkogengar.localdomain kernel: clocksource: kvm-clock: mask: 0xffffffffffff>
Sep 29 13:08:06 rishardkogengar.localdomain kernel: tsc: Detected 3187.202 MHz processor
Sep 29 13:08:06 rishardkogengar.localdomain kernel: e820: update [mem 0x00000000-0x000000fff] usab>
```

Итоги работы

В ходе работы были изучены механизмы регистрации и анализа системных событий в Linux с использованием **rsyslog** и **systemd-journald**. Настроено перенаправление сообщений Apache в системный журнал, организация отдельных логов для ошибок и отладки, а также фильтрация и выборка событий с помощью **journalctl**.